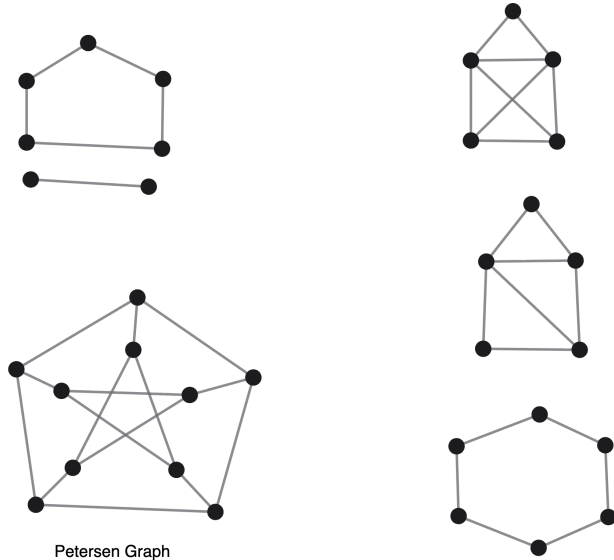


1. Find the chromatic number of the following graphs.



2. Let K_n be the graph on n vertices containing all possible edges (complete graph). What is $\chi(K_n)$?
3. Let C_n be the cycle graph on n vertices. What is $\chi(C_n)$? Prove a formula in general for even and odd n .
4. Let $G = (V, E)$ be a 2-colorable graph.¹ Show that $|E| \leq |V|^2/4$.
Hint: Write $|V| = a + b$ where a and b are the number of red/green vertices. Prove and use the inequality $2ab \leq a^2 + b^2$.

¹These graphs are also called *bipartite*